SPS-24G-01

SPS-24G-01 — Smart Presence Sensor Switch

24 GHz mmWave Automatic Light Switch · Standalone (No Wi-Fi / No Bluetooth)

SAMAY ELECTRONICS · Preliminary Datasheet · Rev A0 · 2026-10-25

> ⚠ **MAINS VOLTAGE PRODUCT.** This datasheet describes an engineering design. The board carries 85–265 VAC on its primary side. It must be reviewed by a qualified engineer and type-tested (IEC 60669 / IEC 61058, BIS IS 60669, creepage/clearance per IEC 60664-1, HIPOT, EMC) before any unit is energised or sold. See `SAFETY_REVIEW.md`.

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1. General Description

The **SPS-24G-01** is a self-contained automatic light switch that replaces a standard wall switch in an Indian 86 × 86 mm modular box. A **24 GHz millimetre-wave radar** (Hi-Link HLK-LD2410B) detects a human — moving *or* completely still — and an on-board **ATtiny13A** microcontroller switches the lamp through a **zero-cross opto-triac + BTA16-600B triac**. When the room empties, an adjustable timer (10 s … 10 min) turns the lamp off automatically. No app, no cloud, no radio pairing — it works out of the box.

Unlike PIR switches, the mmWave radar keeps the light **on while a person sits or stands still** (e.g. on a toilet, in a shower), eliminating the classic "lights-off-while-I'm-not-moving" problem. The radar also sees **through the plastic front cover**, so there is no visible lens.

Key Features

- 24 GHz FMCW presence radar; detects motion and micro-motion (breathing), range up to 6 m, ±60° coverage.
- Universal mains input **85–265 VAC, 50/60 Hz** via isolated HLK-PM01 5 V supply.
- Silent **zero-cross TRIAC** switching (no relay click, long life, low EMI).
- Adjustable OFF delay **10 s – 10 min** (on-board trimmer).
- **Manual override** button (Auto / Forced-ON / Forced-OFF) and a status/fault LED.
- Reinforced **primary↔secondary isolation** (HLK-PM01 + 7.5 kVrms opto), ≥ 6 mm creepage barrier + routed slot.
- Input protection: slow-blow fuse, MOV surge clamp, NTC inrush limiter, X2 EMI cap + bleeder.
- 70 × 56 mm 2-layer FR-4, fits the standard 86 × 86 modular back box.

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2. Absolute Maximum Ratings

(Stresses beyond these may cause permanent damage. Not continuous operating conditions.)

Parameter	Symbol	Min	Max	Unit
Mains input voltage	V_AC	—	265	VAC rms
Mains frequency	f	45	65	Hz
Triac repetitive off-state voltage (BTA16-600B)	V_DRM	—	600	V
Load current through triac (with adequate heatsink)	I_T(RMS)	—	16	A
Load current, **no heatsink** (this board)	I_T	—	~2	A
Fuse rating (F1)	—	—	0.5 (T)	A
MCU / logic supply	+5V	-0.3	5.5	V
Operating ambient	T_A	-10	+55	°C
Storage temperature	T_STG	-40	+85	°C

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3. Recommended Operating Conditions / Electrical Characteristics

Parameter	Conditions	Min	Typ	Max	Unit
Mains input	—	85	230	265	VAC
Standby power (lamp off)	230 VAC, radar active	—	0.4	0.7	W
Isolated rail	HLK-PM01 output	4.75	5.0	5.25	V
Rail current budget	radar 79 mA + MCU + opto	—	110	600	mA
Switched load (resistive/LED)	no heatsink	—	—	2	A
Switched load (with clip heatsink)	TO-220 + sink	—	—	6	A
Minimum load	for reliable triac latch	3	—	—	W
Presence detection range	configurable	0.75	—	6	m
Detection angle	—	—	±60	—	°
Distance resolution	per gate	—	0.75	—	m
OFF delay (no-one timer)	trimmer RV1	10	—	600	s
Turn-on latency (person enters)	—	—	<0.5	1	s
Radar logic OUT level	—	—	3.3	—	V

Note on minimum load: a TRIAC needs a holding current to stay latched. Very low-power LED lamps (< ~3 W) or electronic drivers with tiny input current may flicker or fail to switch cleanly. Fit a snubber (already on board) and, for problem loads, a bleeder/dummy load across the lamp

terminals.

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4. Pin / Terminal Description

4.1 Mains screw terminal — J1 (3-pole, 5.08 mm)

Pin	Name	Direction	Function
1	**L**	Input	Mains Live (Phase) in
2	**N**	Input	Mains Neutral (feeds the power supply)
3	**LAMP**	Output	Switched Live to the lamp. The lamp's other wire goes to Neutral.

> The switch is a **neutral-required** design: the HLK-PM01 supply is powered from L+N so the electronics work even when the lamp is off. Both N and the lamp-return Neutral come from the same mains neutral.

4.2 Radar module header — J3 (1×5, 1.27 mm, vertical)

Pin	Name	Function
1	OUT	Presence output, 3.3 V logic. **HIGH = human present**, LOW = vacant → MCU PB4
2	TX	LD2410B UART transmit (config/debug) → J4
3	RX	LD2410B UART receive (config) → J4
4	GND	Ground
5	VCC	+5 V supply to the radar

4.3 ISP programming header — J2 (2×3, 2.54 mm, standard AVR)

`1 MISO · 2 VCC(+5V) · 3 SCK · 4 MOSI · 5 RESET · 6 GND` — program with **mains disconnected**.

4.4 Radar config UART — J4 (1×3, 2.54 mm)

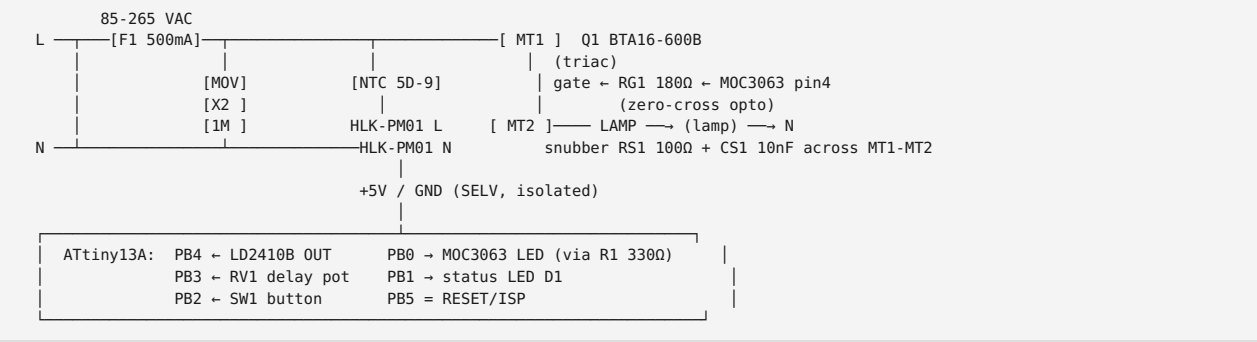
`1 TX · 2 RX · 3 GND` — connect a 3.3 V USB-UART adapter and the Hi-Link PC tool to set detection range, per-gate sensitivity and the radar's own "no-one duration".

4.5 Controls / indicators

- **SW1** — momentary push button: short press toggles Auto ↔ Manual; long press forces lamp OFF.
- **D1** — status LED: solid = presence, slow blink = manual mode, fast blink = fault (radar not responding).
- **RV1** — 10 kΩ trimmer: sets the auto OFF delay (10 s fully CCW … 10 min fully CW), read by the MCU ADC.

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5. Theory of Operation



Sequence

1. **Person enters** — radar OUT goes HIGH → MCU drives the opto → triac fires at the next mains zero-crossing → **lamp ON** (< 0.5 s, silent).
2. **Person stays (even still)** — radar continues to report presence (micro-motion/breathing) → lamp stays ON; the OFF timer is held reset.
3. **Person leaves** — radar OUT goes LOW → MCU starts the OFF delay (set by RV1). If no presence returns before the delay expires, the opto is released and the triac turns off at the next current zero → **lamp OFF**.
4. **Manual override** — a button press forces ON or OFF regardless of the radar; another press (or a timeout) returns to Auto. On power-up the firmware initialises to a safe **OFF, Auto** state.
5. **Fail-safe** — if the radar stops responding, the firmware blinks the LED and defaults the lamp OFF (configurable to last-known-state).

The **MOC3063 is a zero-crossing opto-triac**: it only triggers the main triac near the mains zero-crossing, which minimises inrush, EMI and switching stress — this is why silent, flicker-free switching is possible.

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6. Mechanical

Item	Value
PCB size	**70 × 56 mm**, rounded corners R3
Thickness	1.6 mm
Layers / copper	2-layer FR-4, 1 oz
Finish	ENIG

Mounting holes	4 × Ø3.2 mm (M3), 4.0 mm from corners
Radar mounting	vertical daughter-board on J3, antenna faces the front cover
Board outline	includes a **routed isolation slot** in the primary↔secondary corridor
Enclosure	fits standard Indian 86 × 86 modular back box (internal cavity ≈ 75 mm)

> **Design note on size:** 45 × 45 mm was the preferred target, but the mandated through-hole mains parts (HLK-PM01 34 × 20 mm, 14 mm MOV, TO-220 triac, 5.08 mm terminal) plus a ≥ 6 mm isolation barrier do not physically fit that area. 70 × 56 mm is the compact result that still drops into the 86 × 86 box. To reach ~50 × 50 mm, substitute a smaller supply (e.g. 3 W SMD module), a 10 mm MOV and an SMD triac — see `DESIGN_REPORT.md`.

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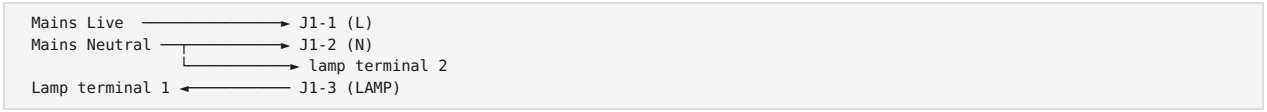
7. Isolation & Safety Summary

- **Primary (mains) section** and **secondary (SELV 5 V) section** are kept on opposite halves of the board, separated by a ≥ 6 mm no-copper corridor with a milled slot.
- The only parts crossing the barrier are the **HLK-PM01** (reinforced AC-DC isolation) and the **MOC3063** (7.5 kVrms opto) — both certified isolation devices.
- Primary-side live-to-live (L–N–Lamp) spacing is set by routing and the component packages; conformal coating is required where it is below 2.5 mm. Full creepage/clearance analysis is in `SAFETY_REVIEW.md`.
- Board is silk-marked "⚠ **HIGH VOLTAGE — RISK OF SHOCK**" on the primary side.

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8. Application Information

8.1 Typical wiring (ceiling light in a bathroom)



Install the module in the wall box with the radar facing the room through the plastic front plate. Keep **metal objects and foil away from the antenna window**. Mount at 1.1–1.5 m height (wall) or on the ceiling.

8.2 Setting the delay and sensitivity

- **OFF delay:** turn RV1 CW for longer, CCW for shorter (10 s … 10 min).
- **Range / sensitivity:** connect J4 to a 3.3 V USB-UART and run the Hi-Link LD2410B PC tool; set the maximum detection gate (0.75 m per gate, up to gate 8 = 6 m) and per-gate sensitivity to reject movement outside the room (e.g. a corridor behind a thin wall). Settings are stored in the radar.

8.3 Load guidance

- **LED lamps:** use the on-board snubber; for < 3 W drivers that flicker, add a 0.1 µF X2 + 100 kΩ bleeder across the lamp, or use the relay-output variant.
- **Resistive / halogen:** fine up to ~2 A without a heatsink; fit a TO-220 clip heatsink for 2–6 A.
- **Inductive (fans/transformers):** keep the snubber; verify dv/dt; prefer a snubberless triac (BTA16-600BW) for heavily inductive loads.

8.4 Firmware

The ATtiny13A firmware implements power-on init, presence read, the adjustable OFF timer, manual override and fail-safe OFF. Pin map and build/flash instructions are in `../06_firmware/`. Program via J2 with an AVR ISP programmer, **mains disconnected**.

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9. Ordering Information

Part number	Description
SPS-24G-01	Smart Presence Sensor Switch, 24 GHz, triac output, 85–265 VAC, modular form factor
SPS-24G-01-R	Variant with relay output (for difficult LED drivers) — *planned*

Manufactured by **SAMAY ELECTRONICS**. This is a preliminary datasheet for an engineering sample; specifications are subject to change and to safety certification.